



IC202P: Design Practicum Mid-Sem Report

Group 49

Automated feature and defect analysis tool for semiconductors materials

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1 Introduction

As semiconductor technology keeps moving forward, undeterred by the many challenges ahead, defects in SEM images can significantly impact the performance of devices. SEM images often contain noise and artifacts that make defect detection challenging. Traditional methods rely on manual inspection or software that may not provide precise edge roughness quantification. The need for an automated, reliable, and user-friendly tool to extract meaningful roughness metrics from SEM images is crucial for advancing material analysis and quality control.

The objective of this project is to develop a web-based tool that processes SEM images, to classify the images into line and contact spaces images, and calculates critical dimensions (CD), line-edge roughness (LER) and line-width roughness (LWR) for line spaces and calculates average diameter and ellipticity for contact spaces accurately. The tool also displays Power spectral density (PSD) curves for images.

2 List of potential solutions

2.1 Traditional Image Processing

Description:

Traditional Image Processing use rule-based image processing techniques such as edge detection, thresholding, and morphological operations to identify defects in semiconductor materials. These methods analyze high-contrast images to detect anomalies based on predefined rules.

Challenges:

- **Limited Adaptability:** These systems work well for simple and well-defined defect types but struggle with complex or evolving defects.
- **High Maintenance:** Requires frequent parameter tuning when defect characteristics change.
- **False Positives/Negatives:** May fail to detect subtle defects or mistakenly classify normal patterns as defects.

2.2 Deep Learning-Based Defect Detection (CNNs, ViTs, etc.)

Description:

Deep learning-based models, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), are trained on large datasets of semiconductor images to automatically classify and localize defects. These models learn feature representations, making them more robust to variations

in defect appearance.

Challenges:

- **Data Dependency:** Requires a large, well-labeled dataset of SEM images for effective training.
- **Computational Cost:** Training and deploying deep learning models require high-end GPUs or TPUs, increasing infrastructure costs.
- **Generalization Issues:** Model performance may degrade when applied to new defect types not present in the training data.

2.3 Unsupervised Anomaly Detection (Autoencoders, GANs, etc.)

Description:

Instead of requiring labeled data, unsupervised learning methods like Autoencoders and Generative Adversarial Networks (GANs) learn the normal distribution of defect-free semiconductor images. Any deviation from this normality is flagged as an anomaly.

Challenges:

- **Subtle Defects:** Some minor defects may not be distinguishable from natural variations, leading to misclassification.
- **Model Sensitivity:** Small environmental variations (e.g., noise in SEM images) may trigger false alarms.
- **Interpretability:** Hard to explain why a specific region was classified as defective.

2.4 Hybrid AI Approaches (Combination of CNNs + Rule-Based Methods)

Description:

This approach combines deep learning-based feature extraction with traditional machine vision rules. CNNs detect complex defect patterns, while rule-based methods refine detection for well-understood defect types.

Challenges:

- **Integration Complexity:** Requires expertise in both deep learning and classical computer vision.
- **Computational Overhead:** Hybrid models can be resource-intensive.
- **Scalability:** Adding new defect types requires updating both the rule-based system and the deep learning model.

2.5 Statistical & Mathematical Approaches

Description:

SEM provides high-resolution images, making it ideal for detecting nanoscale defects in semiconductor wafers. AI models analyze SEM images to classify defects with high accuracy.

Challenges:

- **Assumption Dependency:** Many statistical models assume specific defect distributions, which may not always hold in real-world scenarios.
- **Limited Spatial Context:** These methods may not capture complex spatial relationships between defects as effectively as deep learning models.
- **High Computational Cost for Large Data:** Some techniques, like Fourier or wavelet analysis, require significant computational resources for high-resolution images.

2.6 AI-Enhanced Traditional Image Processing

Description:

Integrates AI with conventional image processing (e.g., edge detection, thresholding) to improve defect detection.

Challenges:

- **Feature Sensitivity:** AI may misinterpret noise as defects, leading to false positives.
- **Lack of Generalization:** AI models trained on specific defect types may not work well on new materials or process variations.

2.7 Integration with MATLAB

Description:

MATLAB provides a powerful environment for signal processing, image analysis, and computational modeling, making it a valuable tool for semiconductor defect analysis. Using MATLAB, we can apply advanced filtering techniques, Fourier transforms, and statistical analysis to identify and classify defects. It also allows seamless integration with machine learning and deep learning models for improved defect detection.

Challenges:

- **Computational Cost:** MATLAB's built-in functions can be computationally intensive, especially for large datasets.

- **Licensing Issues:** MATLAB requires a paid license, making it less accessible for small-scale projects.
- **Integration Complexity:** Connecting MATLAB with external systems may require additional toolboxes and customization.

2.8 Generating SEM Images Using CNN

Description:

Convolutional Neural Networks (CNNs) can be used to generate realistic SEM images of semiconductor materials by learning patterns from existing high-resolution microscopy images. Then it can be used to calculate Line edge roughness and Line width roughness.

Challenges:

- **Data Requirements:** CNNs require a large dataset of SEM images for effective training.
- **Computational Cost:** Training deep generative models is resource-intensive and may require high-end GPUs.
- **Realism & Accuracy:** The generated images must accurately reflect real defects and material properties, requiring domain-specific constraints.

2.9 Diffusion Models

Description:

These models iteratively refine noise to generate high-quality SEM images, often outperforming GANs in fidelity. Stable Diffusion or other text-to-image diffusion models can be fine-tuned for SEM image generation.

Challenges:

- Computationally expensive and requires long training times.
- Needs large datasets to learn meaningful defect representations

2.10 Adaptive Thresholding for Edge Detection

Description:

Use adaptive thresholding techniques to detect edges in SEM images more accurately. Instead of using a fixed threshold, adaptive thresholding adjusts based on local image intensity, improving roughness detection in varying contrast conditions.

Challenges:

- Requires fine-tuning for different SEM image conditions.
- May struggle with noisy images if not pre-processed properly.

2.11 Fast Fourier Transform for Roughness Analysis

Description:

Use Fast Fourier Transform to analyze frequency components of edge variations in SEM images. This method helps in identifying periodic defects or systematic variations in Line edge roughness and Line width roughness.

Challenges:

- Requires pre-processing to remove noise from SEM images.
- Fast Fourier Transform alone does not work well for random roughness variations without additional processing.

2.12 Self-Supervised Learning for Defect Detection

Description:

Self-supervised learning (SSL) reduces the need for labeled data by learning representations from unlabeled SEM images.

Challenges:

- Hard to Fine-Tune for Specific Defects: SSL models may require additional supervised fine-tuning for specific semiconductor defects.
- Computationally Intensive: SSL methods require large-scale training on high-performance GPUs.
- Validation Complexity: Requires extensive validation to ensure that learned representations are meaningful for defect detection.

3 Statistical analysis of the solutions

A: Cost of the Idea ($30000/\text{Amount}$) $\times 10$

B: Availability of Parts

C: Ease of Manufacturing / Assembly

D: Fulfilment of Purpose

E: Ergonomics / Aesthetics

F: Integration of Power Source

G: Integration of Electronics

H: Timebound Manufacturing Feasibility

I: Maintenance / Servicing Cost

J: Overall Feel / Confidence in the Solution

K: Overall Score (Total Out of 100)

Sr No.	Solution Description	A	B	C	D	E	F	G	H	I	J	K
1	Traditional Image Processing	12	10	8	6	10	10	10	8	7	7	88
2	Deep Learning-Based Defect Detection	5	10	9	7	10	10	10	7	5	7	80
3	Unsupervised Anomaly Detection	10	10	10	6	10	10	10	7	6	7	86
4	Hybrid AI Approaches	12	10	10	10	10	10	10	9	8	9	98
5	Statistical & Mathematical Approaches	12	10	10	5	8	10	10	6	5	5	81
6	AI-Enhanced Traditional Image Processing	12	10	10	5	10	10	10	8	4	6	85
7	Integration with MATLAB	4	10	10	7	6	10	10	8	4	7	76
8	Generating SEM Images Using CNN	5	10	10	7	10	10	10	6	4	5	77
9	Diffusion Models	4	10	10	5	10	10	10	6	6	6	77
10	Adaptive Thresholding for Edge Detection	12	10	10	8	10	10	10	8	6	7	93
11	Fast Fourier Transform for Roughness Analysis	10	10	10	7	10	10	10	6	6	6	75
12	Self-Supervised Learning for Defect Detection	9	9	10	8	10	10	10	9	6	5	86

4 Detailed Approach Of Our Solution

Our solution leverages a Hybrid AI approach that integrates traditional image processing, statistical methods, and deep learning to achieve high accuracy, adaptability, and automation in semiconductor defect analysis.

4.1 Data Acquisition & Preprocessing

Use Scanning Electron Microscopy (SEM) images for nanoscale defect detection and calculate the line edge roughness and line width roughness. Next we reduce noise by applying Gaussian /Median filtering to remove noise and apply normalization to scale pixel intensities to a consistent range for deep learning models.

4.2 Edge detection

Using of canny / sobel edge detection with AI-based adaptive thresholding or other methods can be used like edge fit function (liner, polynomial) to detect edges.

4.3 Metrology

After the preprocessing and edge detection, we then measure the line width roughness and line edge roughness for the edges and calculate the other parameters like average critical dimensions, pitch, critical dimensions deviation, power spectral density. For contact specific images, average ellipticity, average diameter, horizontal and vertical pitch calculations.

5 Benchmarking of the finally selected solution

In semiconductor metrology, various software solutions offer distinct features tailored to specific needs. Our AI Hybrid-Based Semiconductor Defect Analysis Tool is equivalent to the all existing metrology applications and provide a competitive performance because:

- **Free to use:** Unlike other metrology applications which are expensive, our website is free to use and does not require any license to purchase.
- **No third party installation:** Our website does not need any application to use the website. Other meterology application required extensive applications like MATLAB to run there applications.
- **Simple:** Our website is designed with a clean, intuitive interface, ensuring effortless navigation and seamless usability, allowing users to analyze semiconductor defects and LER/LWR measurements with minimal effort and maximum efficiency.

6 Website Design

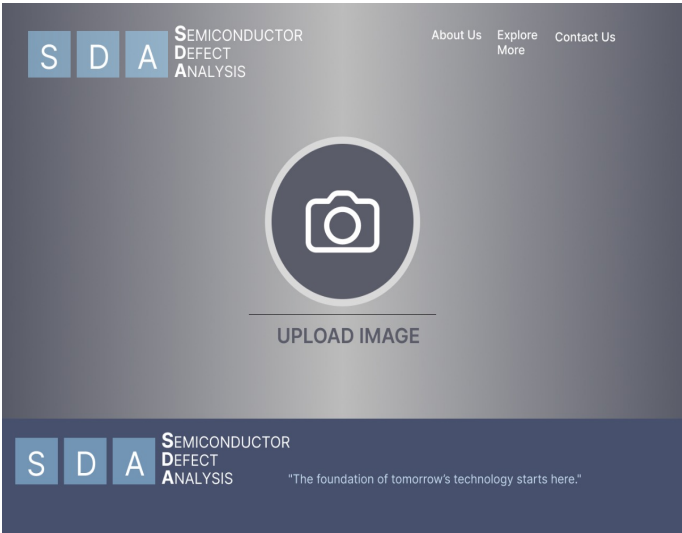


Figure 1: Website Interface

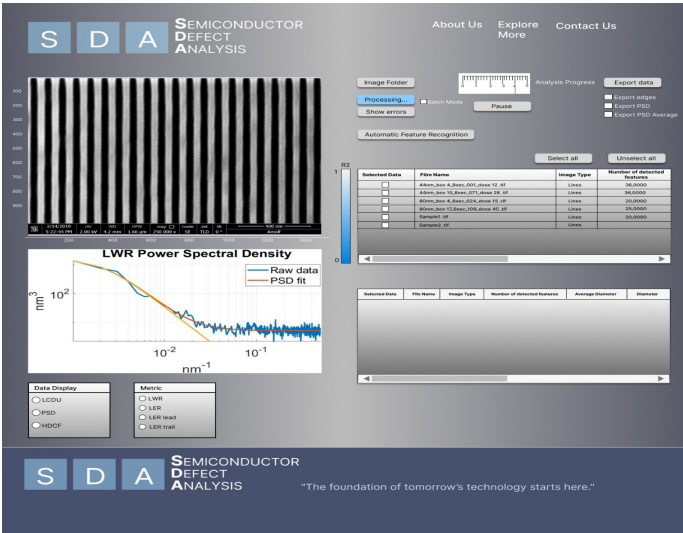


Figure 2: SEM Images Analysis

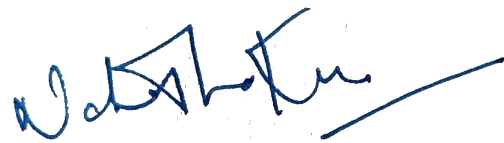


Figure 3: Website Frontpage

7 Bill of Materials

Item Name	Quantity	Cost (in ₹)	Vendor
Website Domain name	1	₹ 8000	Namecheap
Webhosting	1	₹ 16000	Render

Total Cost of the Items: ₹ 24000



Signature of Faculty Mentor

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